

DIT Ion Migration Calculator Quick Start

A concise guide for accessing the public web calculator, uploading DIT data, reviewing the plot, and exporting results.

Access

Open the calculator in any modern browser:

<https://flabio121.github.io/ion-migration-dit-calculator/>

The calculator runs in the browser. Uploaded files are processed locally on the user's computer.

The screenshot shows the DIT Ion Migration Calculator interface. At the top, there's a title 'DIT Ion Migration Calculator' and two buttons: 'Sample' and 'Export results'. Below the title is a subtitle: 'Upload a DIT text file and calculate mobile ion concentration, conductivity, and mobility.' The main area is divided into several sections. On the left, there's a dashed box labeled 'Choose or drop DIT data' with 'No files selected' below it. To the right of this are input fields for 'Thickness' (5.5e-7 m), 'Area' (9e-6 m²), 'Preload bias' (0.8 V), 'Built-in potential' (1.2 V), 'Thermal voltage' (0.026 V), and 'Relative permittivity' (24.2 εr). Below these is a section for 'Ionic resistances' with a '+' button and eight input fields for Device 1 through Device 8, each with a unit symbol Ω. At the bottom, there's a summary section with three columns: 'MEASUREMENTS' (0), 'PROCESSED POINTS' (0), and 'BIAS BRACKET' (--). Below this is a 'DIT plot' section with 'No data loaded' and two buttons: 'Processed' and 'Raw'. The plot area is empty, with a message 'Upload DIT data to view the plot' at the bottom.

Before You Start

Prepare one or more DIT text files. The current calculator expects paired time/current data exported in the same format used by the validated workflow. Have these values ready if needed:

- Device thickness and active area.
- Preload bias, built-in potential, thermal voltage, and relative permittivity.
- Optional ionic resistance values for conductivity and mobility.

Use The Calculator

1. Open the website and drag one or more `.txt` files into the upload area, or click the upload area to select files.

2. Confirm the device constants. Defaults are pre-filled, but should be changed if the device geometry or assumptions differ.
3. Enter ionic resistance values when conductivity and mobility are needed. Leave them blank if only concentration is required.
4. Inspect the DIT plot. Use **Processed** to see the trace used for integration and **Raw** to check the imported data.
5. Review the result table. Key outputs include integrated charge, concentration in cm^{-3} , conductivity, mobility, and data-point count.
6. Click **Export results** to download the summary table as a CSV file.

Notes

- Results depend strongly on active area, thickness, voltage assumptions, and the processed integration window.
- Conductivity and mobility require ionic resistance inputs; concentration does not.
- For reproducibility, keep the uploaded raw data, exported CSV, and device constants with the experiment record.